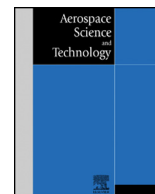




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Electric propulsion system sizing methodology for an agriculture multicopter

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ARTICLE INFO

Article history:

Received 29 March 2018

Received in revised form 13 February 2019

Accepted 25 April 2019

Available online xxxx

Keywords:

Multicopter design

Electrical propulsion sizing

Blade elements momentum theory

UAV

ABSTRACT

Using a suitable propulsion system for Drones capable of obtaining desired performance is one of the primary goals for designing electric multicopters. Choosing the right components for the electric system: Motor, ESC, battery, rotor (propeller) is always challenging because of the diversity of and misleading information on these products on the market. By studying specifications brought by manufacturers and experimental data, the relationship of the components in the propulsion system is given and parameterized. This paper presents a method to size the electric propulsion system for design and manufacturing a multicopter that is able to handle mission requirements. The sensitivity of design parameters to GTOW is given. An Agriculture Quadcopter is built and compared to the results of sizing and the difference stays within $\pm 4\%$ for gross take-off weight. The maximum thrust over weight ratio is introduced, and the cut of payload during operation of pesticide spraying is also accounted in the calculation for improvement of sizing the Quadcopter.

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1. Introduction

Unmanned air vehicles (UAVs) are divided into many types, which can fly a significant range or limit spaces based on their objectives. They can have multiple configurations: fixed wing, rotary wing, etc. The multi-rotor copters (or multicopters) have more advantages than others do: simpler in mechanism than helicopters, highly flexible, easy to control, safer and able to take-off and land vertically. With the advance of technologies such as more efficient motors, higher energy density batteries, faster and smaller flight controller, multicopters are widely used now for various applications in life besides pleasure, such as monitoring environment [1–3], soil erosion [4], radiological inspection [5], inspecting fruit orchards and vineyards [6], search and rescue missions, firefighting and others. Moreover, multicopter is a popular platform for researchers applied and validated their model, control algorithms [7,8] and new ways of power transfer [9–11]. Therefore, an electric propulsion system sizing methodology is needed to be developed for quick components choosing and constructing a new multicopter. This paper presents a design and development of an electric multicopter for agriculture application, the vehicle is mainly used for pesticide spraying to reduce the labor workforce and eliminate the health risk from hazardous plant protection products or taking aerial pictures to evaluate crops and forest conditions. A method

for conceptually sizing the electric propulsion system is also developed and applied for the design of a multicopter that is able to handle heavy payload or long flight time in order to meet mission requirements. A configuration of four actuators (Quadcopter) was chosen for its simple control scheme and a small fraction of structure weight.

Choosing the components for the propulsion system of an electric multicopter (Motors, ESCs, batteries, rotors (propellers)) used to depend on experiences of designers [12] and from the results of estimated performance by applying basic momentum theory [1, 6, 12, 13]. Nowadays, tools as eCalc [14] is popular for doing the task. The list of options will be provided and users can choose the initial components from the available database. From that, the program will calculate and provide parameters such as flight time, power required, performance... However, this program calculates and provides results from the manual selections. Researchers as Abarca have been using this tool for predicting the performance of their multicopters [1], Shi [15] and Kim [16] have verified results from their evaluation methodologies by it. Test bench measurements for motors and propellers are also used for simple sizing of propulsion system [11, 13] or validating results from proposed method [15].

The methodologies that were presented by Bershadsky [17] and Winslow [18] to optimize the gross take-off weight (GTOW) of UAVs using multiple rotors to satisfy flight requirements, but they are applied and confirmed just for small, lightweight quadcopter [19]. These studies present procedures to choose components for

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<https://doi.org/10.1016/j.ast.2019.04.044>

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Nomenclature

a	Angular induced factor	t	Freefall time, s
A	Rotor disk area, m ²	T	Thrust, N
b	Reserve factor	P	Power, W
c	Chord blade, m	v	Specific velocity, m/s
C	Battery capacity, Ah	V	Voltage, V
C_L	Lift coefficient	ϕ	Inflow angle of attack
C_D	Drag coefficient	κ	Technology factor
C_T	Rotor thrust coefficient	θ	Pitch angle at the mid-span position of the blade segment, degree
C_P	Power coefficient	η	Efficiency
C_Q	Equivalent torque coefficient	ρ	Density of the air, kg/m ³
D	Rotor diameter, in	σ	Solidity
D_f	Parasitic drag, N	λ	Inflow ratio
k	Air resistance, kg/m	W	Weight, N
R	Rotor radius, m		
f	Load factor		
F	Prandtl tip-loss factor		
FT	Flight time, min		
I	Current, A		
K_v	Speed constant, RPM/V		
m	Mass of component, g		
N	Number of components or rotation speed, RPM		
R	Resistance of component, Ohm		
r_n	Radius at the mid-span of each segment, m		

Subscripts

O	No-load
M	Motor
max	Maximum continuous
b	Battery
$props$	Propeller
ESC	Electronic speed controller
PA	Payload and avionics

propulsion system which makes the lightest final GTOW or maximizes endurance [16]. The optimization method is also applied to create a design for multicopters whose constraints are energy consumption, length, size of components and others depending on objectives [20] and a genetic algorithm was used for multicopter design optimizing [21]. Some of the methodologies for the preliminary design were based on statistical data available from manufacturers [17,18,22]. From that, sensitivity analysis for the UAV's endurance and take-off weight is presented to aid the application of Multidisciplinary Design Optimization (MDO) to optimize the electric propulsion system for UAV by Gur and Rosen [23] and a design optimization process of multicopters was used for evaluating their configuration (quadcopter, hexacopter and octocopter) by Magnussen [24]. Tyan [25] introduced a sizing and resizing method that helped to select parts of the propulsion system and predicted the parameters of a 3.5 kg hybrid electric UAV.

In this study, for designing an electric propulsion system for a heavy lift multicopter carrying a heavy payload with specific flight requirements, an algorithm is written to obtain a minimal GTOW. From there, the masses of components are parameterized by the characteristics of the propulsion system to help select appropriate components to ensure that the flight requirements are met. Since the expected GTOW is obtained, the designers can proceed to predict their performance and then they can design suitable components such as rotors, motors, speed regulators (ESCs) and batteries. By that, the goal of this study is conceptually designing an electric propulsion system for an agriculture application multicopter and a pesticide spraying Quadcopter is constructed and tested in reality. Load factor f (maximum thrust over weight ratio) for preliminarily designing multicopters is discussed in a new approach. Furthermore, this paper also presents the calculation for the effect of payload mass that changes over time based on mission profile, and it accounted for forward flight performance.

2. Propulsion system sizing methodology for multicopters

The input of the algorithm includes the parameters of the attached devices (power and mass of the payload and avionics:

P_{PA}, m_{PA}), flight conditions (total flight time FT , maximum use of available thrust ATP , load factor f), environmental conditions (temperature T , pressure p , etc.), propeller parameters (predicted diameter D_{in} , Cl_α , C_{d0} , number of blades N_b , etc.), motor parameters (predicted motor efficiency η_M , number of motors N_m , etc.), ESC efficiency η_{ESC} , battery (number of cell S , reserve factor b , etc.). The input parameters are the user's requirements depending on mission profiles, working conditions of the multicopter, as well as based on experiences. In the future, design optimization methods could be applied for multiple parameters to achieve highest efficiency, endurance, range or load-carrying capacity and other performance requirements. In this study, the objective of the sizing process is generating a propulsion system for the multicopter which will allow a vehicle to accomplish a specific mission with the smallest possible GTOW.

The algorithm is a loop with an initial gross take-off weight (GTOW_i) and other input parameters. From the initial GTOW, the required thrust for each motor could be determined by the equilibrium of forces in operation cases such as hover or maximum thrust conditions as discussed in section 5.2. The predicted propeller diameter will be run through a loop to ensure that the Mach number at the blade tip M_{tip} is not near the critical Mach number $M_{critical}$. The M_{tip} is calculated from the predicted rotor diameter D_{in} and the rotation speed of the propeller via equation (25). If the M_{tip} is not satisfied, the rotor diameter will be increased. From the rotor diameter, the modified BEMT presented in section 5 is used for predicting the required performance parameter of the propulsion system such as rotation speed and power from equation (25) and (30), respectively. The sizing algorithm could be used for the simple mission profile with only two stages (hover and maximum throttle) or specific mission profile with the decreasing of payload through time and forwarding flight. By parameterizing the characteristics of the components, their resistances are found from equations (2), (3), (5) and initial parameter also calculated (the K_v of motors from number of cell S and calculated rotation speed, maximum continuous current of ESC and battery's capacity from equations (13) and (15), respectively). After finding out the resistances, the algorithm is able to recalculate the real mean volt-

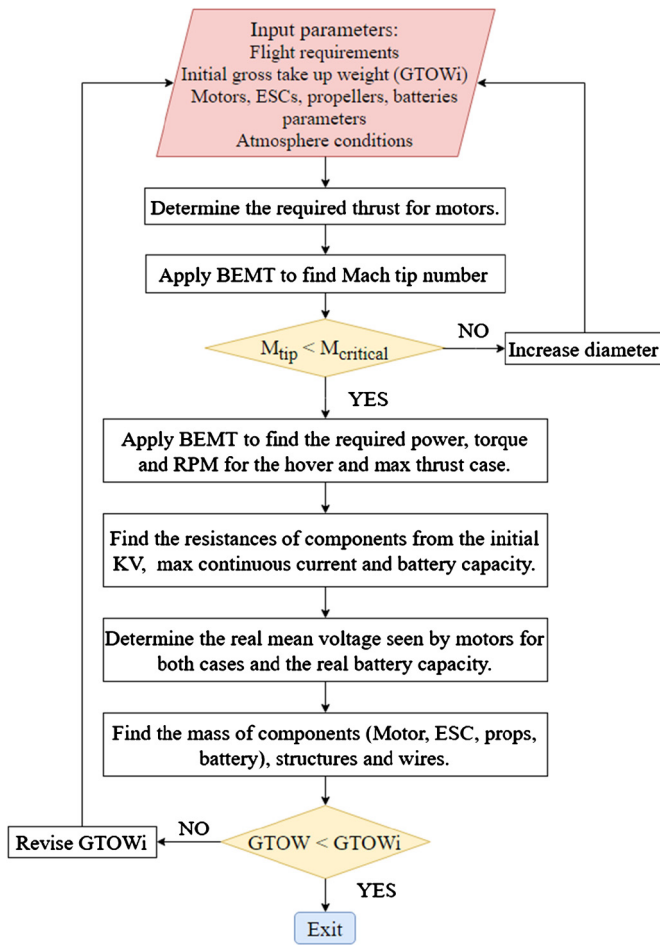


Fig. 1. An algorithm for multicopter propulsion system sizing.

age seen by the motors as shown in equation (12) and also the required battery capacity via equation (15). Finally, the mass of components is calculated from equations (1), (4), (6)–(8), (9) and masses of structure and wire are predicted depending on types of multicopter as discussed in section 3.5. The characteristics trends and relationship between the propulsion system components are discussed in section 3.

The loop ends when the expected GTOW value, which is the sum of components, structures and wires masses, is smaller than the predicted GTOW (Fig. 1). Otherwise, the predicted take-off weight is increased and the loop ends when the minimum weight condition is satisfied. At this time, the configuration of the propulsion system will be given as well as the total take-off weight of the aircraft (final GTOW) is found to meet the requirements of flight operations or design goals.

3. Electric propulsion system

3.1. Motor

An electric Motor is a type of engine that transfers electrical to mechanical energy. There are two types of motor using direct current (DC motor): brushed (BDC) and brushless motor (BLDC). BLDC is usually used for heavyweight multicopters for their low resistance and high-efficiency. The BLDC motors are made of an array of winding and permanent magnets. They are also divided into two types: Out-runner (OR) and In-runner (IR), classified by rotating part. Nowadays, OR BLDC is considered as a better choice than IR, OR motors offer lower Kv (RPM/V) than another type of BLDC, by that, they have low rotation speed and high torque.

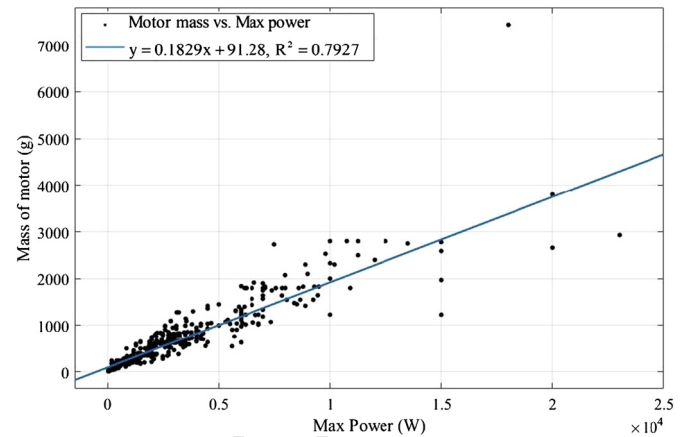


Fig. 2. Motor mass as a function of maximum power (from the data sources [14,26]).

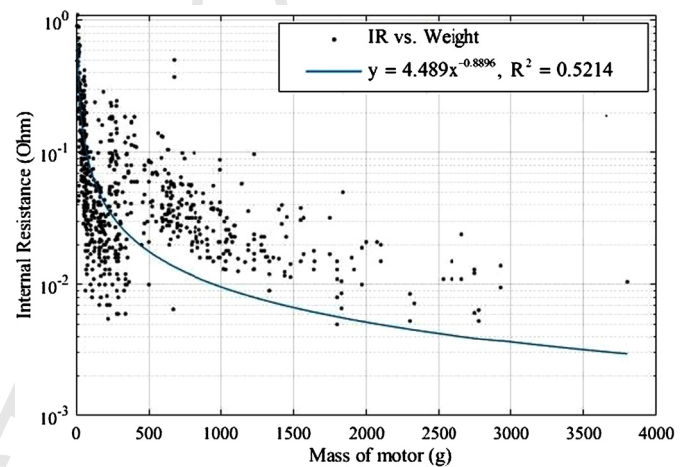


Fig. 3. Motor's internal resistance as a function of mass (from the data sources [14, 26]).

Kv (RPM/V) is speed constant, which will determine the rotation speed of electric motor when no-load and stable voltage was supplied. This is an important element for choosing a motor which is compatible with the power supply and propellers to achieve the requirement speed. If the Kv or the voltage is too high, the motors could be unable to handle the loads at high throttle or rotate in a lower efficiency at low throttle. In general, with large diameter propellers, using a low Kv motor will be more efficient than high Kv motor with small diameter ones (which will provide higher mobility) [17].

The weight of BLDC motors can be parameterized from its maximum power, which is shown in Fig. 2 and as function (1). The bigger motors are, the more torque, output power that motors provide. Furthermore, the motor mass also can be predicted by the size and performance parameters of motors as presented by Winslow [18].

$$m_m = 0.1829P_m + 91.28 \quad (1)$$

The internal resistance ($m\Omega$) of motors, affected by motor winding resistance, is a quantity that will be changed over time with the increment of temperature while operating. Motor's resistance changes are mapped by motor mass as shown in Fig. 3.

The motor's internal resistance expressed as a function of mass:

$$R_m = 4.489m_m^{-0.8896} \quad (2)$$

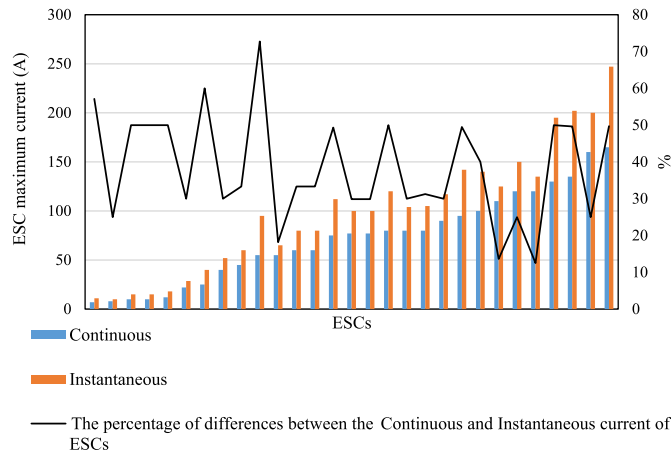


Fig. 4. Comparison of maximum continuous and instantaneous current from various ESCs (from the data source [14]). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

The no-load current (A) of motor is the current when it does not handle any load on its shaft and this parameter is influenced by the motor's internal resistance.

Efficiency is one of the significant parameters and it is not constant through operation time, the relationship between the motor efficiency and its input current is described by equation (14). Usually higher the rated load, higher the efficiency; however, when operating at high load for a long time, the efficiency will drop due to the increment of temperature (which will be increased by the internal resistance and vice versa). With the same type of motor, the operating temperature will be changed over time and it has a great impact on the motor's performance. Motors with high-end technology, manufacturing quality (capable of operating at high load for a long period) will cost more than the same type with lower ability.

3.2. Electronic speed controller

Electronic speed controller (ESC) is a communicator between motor and flight controller. ESC takes the control signal and decides how much power to supply motor by using pulse width modulation (PWM), this technique is utilized to change motor's power with high-efficiency. The efficiency of the system will be decreased due to power losses in the form of heat during operation of transistors and diodes in ESC [27]. The maximum current will be considered which is suitable to the required power for each motor. Maximum allowable continuous current is the current that ESCs can handle for a limit period and max peak current (burst current) effects to the performance of multicopters as how they can be capable of handling the situations that require increasing power suddenly.

By conducting a survey over 55 ESCs from various manufacturers [14], the burst current can vary from 1.1 to 1.7 times greater than maximum allowable current, due to the needs and electrical standards of the manufacturers as shown in Fig. 4. Besides, the operation voltage, input and output signal, internal resistance, etc. should be considered. The ESC's internal resistance is parameterized by its maximum continuous current in Fig. 5.

The ESC's internal resistance shown as a function of its maximum continuous current:

$$R_{ESC} = 0.09619I_{max}^{-0.779} \quad (3)$$

The weight of them is parameterized by their maximum continuous current in Fig. 6.

$$m_{ESC} = \kappa_1 (0.9546I_{max} - 2.592) \quad (4)$$

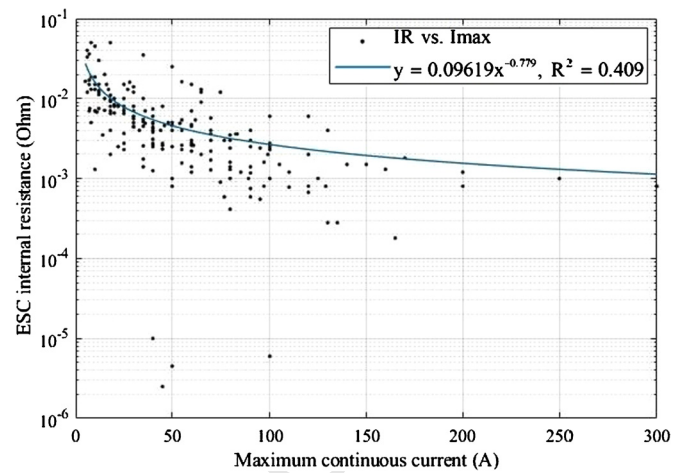


Fig. 5. ESC's IR as a function of maximum continuous current (from the data sources [14,26]).

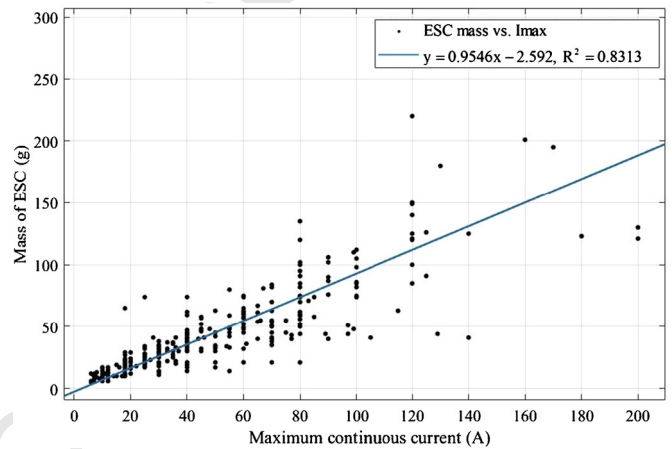


Fig. 6. Mass of ESC as a function of maximum continuous current (from the data sources [26]).

3.3. Battery

A battery is an energy container device that converts chemical energy into electrical energy and vice versa. There are many types of battery which were utilized for multicopters on the market as they were classified by their chemical compositions: Lithium Polymer (LiPo), Lithium Ion (Lilon), Nickel-Cadmium (NiCad), Nickel-metal hydride (NiMH), Lithium iron phosphate (LiFe), etc. With the same battery type, the manufacturer also classifies the batteries in terms of power and capacity. Batteries can have high energy or energy density, but they cannot achieve both. The life cycle of a battery is also an important criterion for classification, with the same capacity and energy, depending on the technology of the manufacturer that gives different discharge or life cycles.

Commonly, batteries which are used for hobby-grade UAVs are the LiPo batteries. With the advantage of high energy density and discharge rate, associated with lightweight due to its high specific energy. More than 90% of small-size unmanned aerial vehicles use LiPo batteries [28]. However, high power density is an advantage which is the reason that Lilon batteries are also a better choice for long endurance, low power or super small multicopters by their lightweight.

Some main parameters must be included besides the chemical composition of batteries. The Cell of a battery determines its output voltage, the nominal voltage (the default, resting voltage of a battery pack) for a LiPo batteries Cell is approximately 3.7 V.

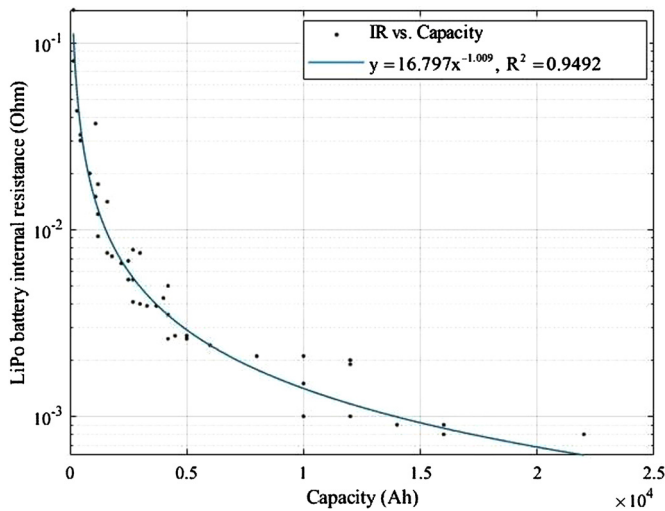


Fig. 7. LiPo battery's internal resistance as a function of capacity (from the data source [14]).

The LiPo battery is considered to be fully charged when it reaches 4.2 V/cell, and will decrease during discharging; 3.7 V/Cell level is average and considered as the rated voltage of a LiPo battery's Cell.

The battery power that the battery can deliver in one hour is battery capacity (mAh). The caution is to use only 80% of the maximum battery capacity to ensure the battery life as it is the stage that falls at the end of life region for Lithium batteries in general. The number can be different depending on a battery's depth of discharge, in other words, it is affected by the voltage profile of the Lithium battery [28]. Battery capacity and mass are influenced by specific energy (Wh/kg). Specific power (W/kg) is the maximum available power per unit of mass which determines the amount of battery weight required to achieve the required power. Both properties depend on the chemical composition of the battery pack. The nominal energy of a battery per unit volume is called the battery energy density (Wh/L) and its power density (W/L) is the maximum capacity of a battery per unit volume. Along with the power consumption and required operating power of the device, the size of the battery pack can be determined. Maximum Continuous Current (A) is the product of the battery capacity and C-rating. Discharge rate (C-rating) is a manufacturer's indication of the discharge capability of the battery. In addition, to ensure electrical safety, battery manufacturers also provide the maximum instantaneous current, which is the current that the battery can withstand for a short period.

The internal resistance of the battery (IR, mΩ) can change over time, depending on the temperature of the battery, the high operating temperature will be equivalent to the high resistance. Typically, batteries with high C-rating give a low resistance. After examining LiPo batteries from Tattu manufacturer [29], the internal resistance parameter in terms of battery capacity is shown in Fig. 7.

From Fig. 7, it is easy to see that the smaller the battery capacity, the greater the internal resistance and it can be approximated with the equation in terms of capacity (mAh):

$$R_b = 16.797C^{-1.009} \quad (5)$$

Connecting the same type of batteries in the series will increase the voltage, but the capacity remains the same. However, when pairing multiple serial cells, it is very easy for a dead cell to cause the whole set to fail. Connecting in parallel will increase the battery capacity but the voltage remains the same. A cell with high internal or error does not exert a serious impact as on the battery

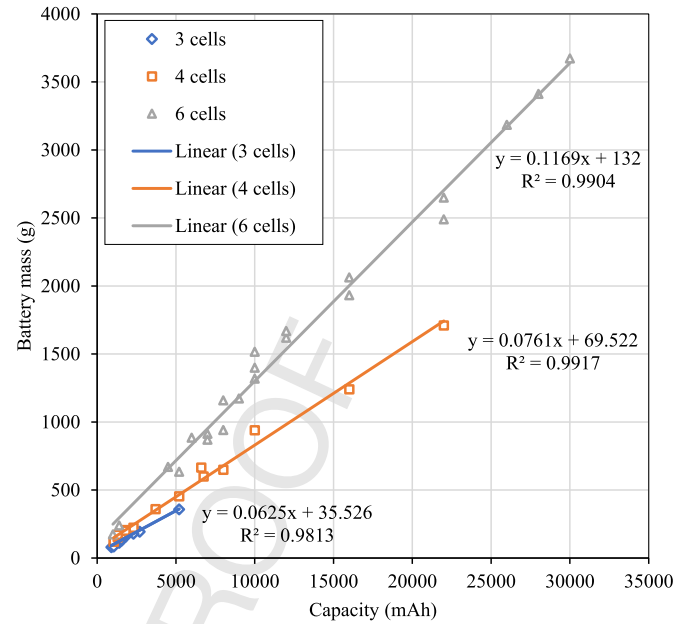


Fig. 8. LiPo battery weight as a function of capacity (from the data source [29]).

pack with serial configuration. However, it will reduce the capacity and total discharge current of the battery [30].

The mass of the LiPo battery can be parameterized by the numbers Cell as shown in Fig. 8. LiPo battery has a corresponding voltage of 3 Cells:

$$m_b = 0.0625C + 35.536 \quad (6)$$

LiPo battery has a corresponding voltage of 4 Cells:

$$m_b = 0.0761C + 69.522 \quad (7)$$

LiPo battery has a corresponding voltage of 6 Cells:

$$m_b = 0.1169C + 132 \quad (8)$$

3.4. Propeller

The propeller (rotor) includes many parameters to consider such as propeller material, diameter, blade shape (chord line and blade twist at each wing position), pitch and number of blades. Currently, on the market, the blades are manufactured with a variety of materials depending on demand and cost: carbon fiber (CF), nylon, plastic and wood. Propeller's material significantly influences its aerodynamic performance: While rotor blades rotate, the angle of attack at each region on the spanwise could change with reference to the original design of the blade [17]. Carbon fiber propellers are known for their stiffness and lightweight but its disadvantage is high cost.

Increasing the pitch as well as the number of blades will provide more thrust, but the propeller efficiency will decrease while the required electrical power and the mechanical power will have to increase as well. Increasing the diameter will rise the efficiency, but the ability to handle the load of the motor must also be increased. When the blades are larger, with other parameters constant, it will rotate at lower revolutions to produce the same lift. At this point, the induce velocity will decrease. Thereby, the efficiency of the system will be raised [17]. Depending on the parameters required during operation such as disk loading, range, flight time that propeller and motor will be selected accordingly.

After examining more than 50 propellers in UIUC database [31], the mass parameters according to their diameter of propellers made of carbon fiber are shown in Fig. 9.

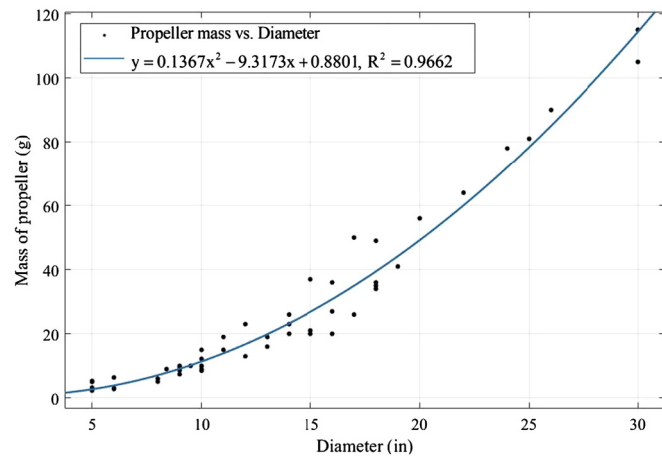


Fig. 9. Carbon Fiber propeller mass as a function of diameter (from the data source [32]).

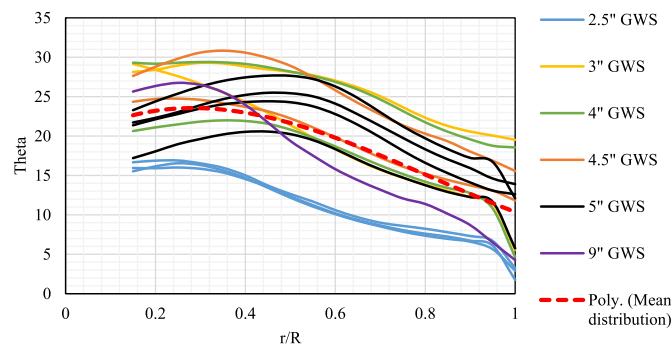


Fig. 10. Blade twist at radial sections r/R for various propellers (from the data source [31]).

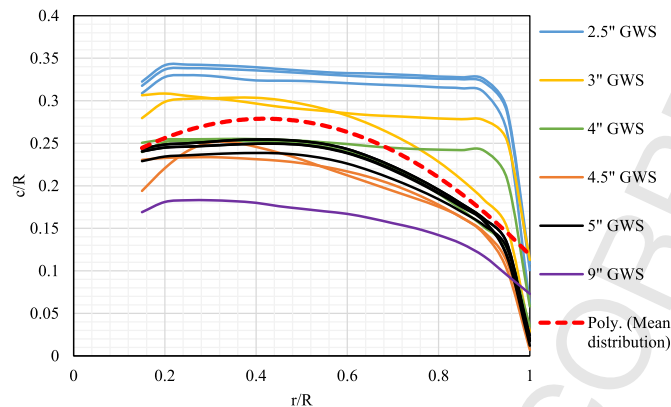


Fig. 11. Wing chord at radial sections r/R for various propellers (from the data source [31]).

$$m_{props} = \kappa_2 (0.1367D^2 - 0.3171D + 0.8801) \quad (9)$$

Depending on which types and manufacturers, a propeller has its blade twist angle distribution as well as the distribution of chord according to the blade segment r/R differently. From the UIUC database [31], surveying large diameter propellers (9 to 11 inches), the distribution of the twist and blade chord is shown in Fig. 10 and Fig. 11, respectively and equations of the corresponding mean values are given:

$$\theta = 29.864 \left(\frac{r}{R} \right)^3 - 72.479 \left(\frac{r}{R} \right)^2 + 33.902 \left(\frac{r}{R} \right) + 19.087 \quad (10)$$

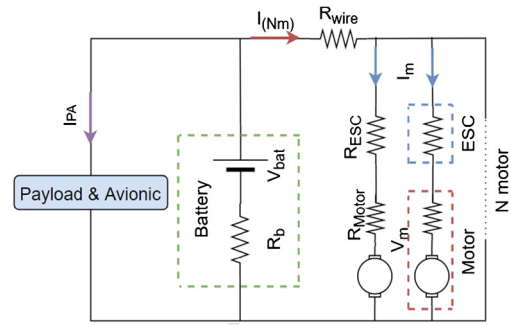


Fig. 12. Multicopter electrical model.

The equation for the mean wing chord along the radius:

$$\frac{c}{R} = -0.4748 \left(\frac{r}{R} \right)^2 + 0.3978 \left(\frac{r}{R} \right) + 0.1956 \quad (11)$$

The propeller is one of the essential parts of a multicopter propulsion system, precisely designing propeller's blade must be taken for the most desired performance achievement. Aerodynamic design optimization of helicopter rotor blades has been applied for hover [33] and forward flight performance [34] and it could also be applied for multicopter rotors to determine better blade's characteristics such as its shape and airfoil shape than the one from the market. Furthermore, an optimization framework for helicopter rotors in hover and forward flight has been created [35] based on high-fidelity coupled CFD/CSM analyses that could be used for multicopter cases for their complex geometries.

3.5. Frames and wires

Typically, for all types of UAVs, propulsion systems account for between 40% and 60% of total take-off weight [28], the rest are frames, wires and other needed accessories. However, different flight requirements and kinds of multicopters that the mass distribution of components in the system will be different. For lightweight ones, the weight of the structure may account for only 8% of GTOW, but for heavy multicopters, it can be up to 40%, and for conventional drone types, that ratio is about 19% on average. Electric cables, which has been found to account for about 5% of the GTOW [17].

4. Electrical model

The propulsion system of a multirotor UAV is a combination of components: battery, a number of motors and ESCs connected in parallel.

From the schematic diagram of the electric multicopter propulsion shown in Fig. 12, the voltage to the motor is calculated:

$$V_m = V_{bat} - I_m N_m (R_b + R_{ESC} + R_m) \quad (12)$$

where I_m is the current in the branch containing the motors, N_m is the number of motor, V_{bat} is the voltage of the battery and the resistance of the battery, ESC and motor.

The input current to the motor can be obtained from the motor's power consumption and it is influenced by the efficiency of the ESCs, motors as well as the no-load current of the motor:

$$I_m = \left(\frac{P}{\eta_M V_{bat}} + I_0 \right) \frac{1}{\eta_{ESC}} \quad (13)$$

The relation between the efficiency and input current of the motor is also expressed as a function:

$$\eta_M = \eta_D \left(1 - \frac{I_m R_m}{V_m}\right) \cdot \left(1 - \frac{I_0}{I_m}\right) \quad (14)$$

An iterative procedure for calculating input current I_m could be obtained by using the Newton-Raphson method for equation (13) and (14).

The total flight time FT is the input of the algorithm which is defined as the sum of hover duration and the time the drone operates at full throttle, determined by the ratio f between maximum thrust and its weight. Capacity of the battery can be found, which is multiplied by the reserve factor b and depend on battery's deep of discharge to ensure battery life:

$$C = (I_{\text{hover}} N_m (1 - f) + I_{\text{max}} N_m (f) + I_{PA}) \times \frac{FT}{60} \times b \quad (15)$$

5. Performance calculation

5.1. Required power

The thrust and power of the rotor are expressed in non-dimensional form with parameters including the induced velocity through the disk rotor, the reference area of the disk rotor A , the velocity at the blade tip V_{tip} and the density of the air ρ . Velocity at the wing tip is calculated: $V_{\text{tip}} = NR$, where N is the radial velocity of the rotor (rad/s) and R is the radius of the propeller (m).

Rotor thrust coefficient:

$$C_T = \frac{T}{\rho A V_{\text{tip}}^2} = \frac{T}{\rho A N^2 R^2} = \frac{T}{\rho \pi N^2 R^4} \quad (16)$$

Power coefficient of the motor:

$$C_P = \frac{P}{\rho A V_{\text{tip}}^3} = \frac{T}{\rho A N^3 R^3} = \frac{T}{\rho \pi N^3 R^5} \quad (17)$$

The equivalent torque coefficient:

$$C_Q = \frac{Q}{\rho A V_{\text{tip}}^2 R} = \frac{Q}{\rho A N^2 R^3} = \frac{Q}{\rho \pi N^2 R^5} \quad (18)$$

By modifying the numerical methods in blade elements momentum theory presented by Leishman [36], the required power can be found. When using this method, the blades are subdivided into many elements along the radius R . Where $n = 1, N$ is the position of the element and $r_n, \theta(r_n)$ are the radius and the twist angle at the mid-span position of each the n segments, respectively. The value of wing coverage (Solidity) is calculated at each of the differential radius of the blade:

$$\sigma = \frac{N_b c(r_n)}{\pi R} \quad (19)$$

The modified methodology is applied for the multicopter since the vehicle operates in a much higher revolution per minute (RPM) than helicopter and wind turbine. The tip vortex generates induced velocity v_i that decreases the local angle of attack α due to strong interaction occurring between the tip vortex and the flow in the plane of the blade, but still higher than those of a helicopter. Therefore, neglect of some parameters in the analysis for a helicopter does not have much accuracy for a multicopter's rotor case. The calculation presented below is considered the effect of wake rotation and applied strip theory.

The following equation will be used for solving the inflow ratio:

$$\lambda(r_n) = \left(r_n \left(4F\lambda_c + \left(\frac{-(C_L(r_n) \cos \phi(r_n) - C_D(r_n) \sin \phi(r_n))^3 \cdot \sigma^2}{+8r_n(C_L(r_n) \cos \phi(r_n) - C_D(r_n) \sin \phi(r_n))^2 F\sigma + 16F^2 \lambda_c^2} \right)^{1/2} \right) \right) \quad (20)$$

With $\phi = \theta - \alpha$ and F is Prandtl's correction factor:

$$F = \left(\frac{2}{\pi} \right) \cos^{-1} \left(\exp \left(-\frac{N_b (1 - r_n)}{2 \lambda(r_n)} \right) \right) \quad (21)$$

Then the inflow ratio will be solved iteratively between equation (20) and (21).

After the inflow was calculated, the increment of thrust coefficient at each blade sections can be obtained:

$$\Delta C_T = 4F(\lambda - \lambda_c) r (1 + a(r_n))^2 \Delta r \quad (22)$$

When a is the angular induced factor, defined by the ratio of angular induced velocity and angular velocity:

$$a(r_n) = \left(\frac{4F \sin \phi \cdot \cos \phi}{\sigma (C_L(r_n) \cos \phi - C_D(r_n) \sin \phi)} - 1 \right)^{-1} \quad (23)$$

From there, the overall thrust of the rotor can be found by integrating the whole blade. Using the simplest approach is the rectangular rule, assuming that the inflow and thrust are constant (constant) at each segment, the thrust coefficient of the rotor can be determined:

$$C_T = \sum_{n=1}^N \Delta C_{T_n} \quad (24)$$

From the thrust coefficient, when considering the hover case with the thrust balanced by the total take-off weights, the revolution of the rotor can be found:

$$N = \sqrt{\frac{T}{\rho \pi R^4 C_T}} \quad (25)$$

The corresponding power coefficient included induce power and profile power:

$$\Delta C_{P_{\text{induced}}} = \frac{1}{2} F \sigma (r_n^2 (1 - a)^2 + \lambda^2) (C_L(r_n) \sin \phi + C_D(r_n) \cos \phi) r_n \Delta r \quad (26)$$

$$\Delta C_{P_{\text{profile}}} = \frac{1}{2} \sigma C_D(r_n)^3 r_n^3 \lambda \Delta r \quad (27)$$

$$\Delta C_P = \Delta C_{P_{\text{profile}}} + \Delta C_{P_{\text{induced}}} \quad (28)$$

$$C_P = \sum_{n=1}^N \Delta C_P \quad (29)$$

Therefore, with the focus on vortex along the blade, the accuracy estimated power consumption is reached, which is assumed lossless by the shaft and gearbox, not included the drive efficiency:

$$P = \rho \pi R^5 N^3 C_P \quad (30)$$

5.2. Maximum thrust to weight ratio

The ratio of maximum thrust and GTOW, also known as the load factor f , also means the ability of a multicopter to lift its own weight and the determinant of its flexibility, capability of withstanding wind gusts, maneuver margin. With a large load factor, the aircraft can quickly recover when lift is suddenly lost or meets an obstacle so the Drone has to perform a sudden emergency climbing and it also depends on the ability to withstand an abrupt increment in the electric current of ESC. Through surveys of several types of multicopters in the market with a take-off mass of more than 1,000 g, this factor is generally 2.3 for small, hobby-grade Drones that require high mobility. For multicopters which

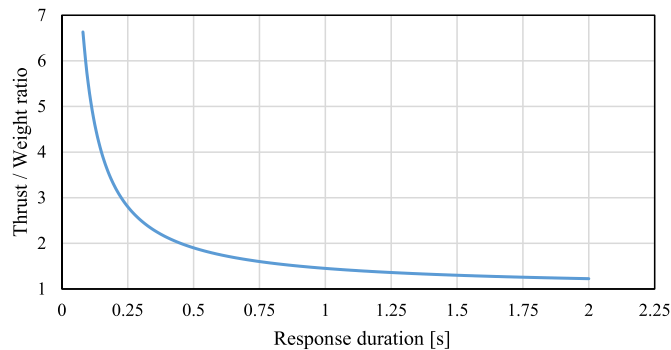


Fig. 13. Thrust over weight ratio depends on response duration for a 10 kg payload designed Agriculture Quadcopter after 1 second of free fall (equivalent 2.95 meters).

are intended for use in environments without high mobility (agriculture, industry), the coefficient may decrease to 1.8, and this ratio can be selected based on experience.

In this paper approach, the load factor f is also studied to select the suitable ratio for designing mission requirements qualified multicopters. This ratio is considered as the capability of restoring the multicopter hover state after an interrupted event which the vehicle loses all of its thrust during operation. The first stage is that the drone falls freely. The vertical drag force acting on multicopter by air resistance is squared by its falling speed and will be presented in the later part. An equivalent wetted area is used:

$$f = \frac{D_f}{\frac{1}{2}\rho V^2} \Leftrightarrow D_f = \frac{1}{2}f\rho V^2 \quad (31)$$

The total force acting on the multicopter at any time t during free fall (without consideration to other factors such as turbulence or wind in any direction) is given:

$$F_{tot} = W - D = mg - \frac{1}{2}f\rho v^2(t) \quad (32)$$

An explicit formula for specific velocity after the duration of free fall time t is:

$$v(t) = \sqrt{\frac{mg}{k}} \left(1 - e^{\left(-\frac{2(\frac{1}{2}f\rho)}{m}\right)t}\right) \quad (33)$$

After a duration of fall time, the controller has to react to achieve hover state for an amount of time (response duration), so counter force needs to be applied. The load factor f is the ratio of that force and the multicopter gross take of weight; it is shown in Fig. 13 that the higher this factor is, the more capable the drone is regarding its ability to recover its hover state.

6. Validation

The mission profile for the drones is hovering all time with full-load for the validation.

6.1. Comparison with DJI-MG1

The algorithm is applied for heavy payload multicopters: Octocopter Agras MG-1 pesticide spray drone [37], due to the specific, full range of specifications given by the manufacturer and on our self-developed Agriculture Quadcopter. The input parameters are listed in Table 1 and for both cases, drive efficiency is fixed and constant at 95% and the ESC efficiency is approximated at 100% due to insignificant heat loss. The retrieved data sources [14,26] collect statistics for a long period in order to provide an abundance of available data and there are differences in production method

Table 1
Input parameters for the methodology.

Input parameter	Unit	Agras MG-1 [37]	Designed Agriculture Quadcopter
Power of payload	W	40	40
Power of avionic	W	10	10
Payload weight	g	10000	10000
Avionic weight	g	386	224
Maximum hover flight time	min	10	15
Propeller diameter	in	21	30
Number of Cell		12	12

Table 2
The results from the methodology tested for Agras MG-1 [37].

Model		Agras MG-1		%
		Calculated	Given by manufacturer [37]	
GTOW	g	21680	22500	-3.64
Battery capacity	mAh	13097	12000	9.14
Maximum continuous current	A	25.3	25	1.2
Max power consumption	W	6252	6400	-2.31
Kv		113.5	130	-12.69
Battery weight	g	3326.1	3700	-10.11
Mass of frames, wires	g	5425	5832	-6.98
ESC weight ^a	g	23.72	55	-56.87
Motor weight ^a	g	234.2	230	1.83
Propeller weight	g	59.95	58	3.36

^a Without cables.

from manufacturers. For the MG-1, the ESC from the manufacturer is much heavier than calculated results because it consists of accessories such as fans and additional electronic circuits for improvement of heat dissipation capability, as well as ability to handle a higher load than the same type of equipment on the market. Moreover, high current ESCs are usually equipped with heat radiator made of metal instead of being wrapped in plastics for electrical insulation like the low current one. Accordingly, a technology factor κ_1 , κ_2 of 1.1 is applied for ESC and propeller's weight, respectively, in both cases to ensure the desired output by studying the results from the application of the methodology and MG-1 specification [37].

From the results shown in Table 2, compared to the manufacturer, the differences of the GTOW from the algorithm and manufacturer are quite small ($\pm 4\%$). The difference of the Kv is quite considerable due to the size of the motor and the aerodynamic properties of the rotors are only estimated, so the precise geometry of the propellers and its calculation model have an important role in the algorithm for their impact on performance calculation precision. Overall, the evaluated results when compared with the MG-1 have shown that the methodology is acceptable for preliminarily sizing a heavy payload multicopter with the acceptable differences of some parameters from estimated and given by manufacturer.

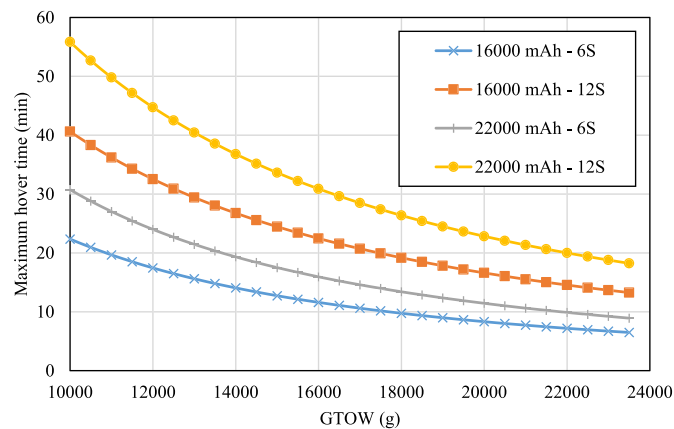
6.2. Applying the methodology for designing an Agriculture Quadcopter

Then the methodology is applied for choosing components to build a multicopter [38] that can be able to carry a heavy payload for a longer time. Overall, the differences from the results and real values are still minor for the Agriculture Quadcopter, besides the weight of separate components from the propulsion system. Because the vast divergence of data was surveyed from numerous fabricators and data sources, their mass varies according to the manufacturer's technologies and material's selections for a given parameter. The methodology also predicts quite accurately the performance for specific input parameters. The gross take-off weight

Table 3

The results from the methodology applied for an Agriculture Quadcopter.

Model		Designed agriculture quadcopter		%
		Calculated	Given by manufacturer and from measurement	
GTOW	g	22663	23041	−1.64
Battery capacity	mAh	16195	16000	1.22
Maximum continuous current	A	74.8	80	−6.5
Max power consumption	W	8433.2	9137.6	−7.71
Kv		98.4	100	−1.6
Battery weight	g	4050	4092	−1.03
Mass of frames, wires	g	5675	5500	3.18
ESC weight ^a	g	75.7	74.2	2.02
Motor weight ^a	g	477	600	−20.5
Propeller weight	g	125.83	132	−4.67

^a Without cables.**Fig. 14.** Calculated maximum hover time vs. gross take-off weight: The figure shows the estimated hover time of the Agriculture Quadcopter for several of batteries configurations.

of the 10 kg payload designed Agriculture Quadcopter could still be decreased because the structure of the quadcopter is not yet optimized (Table 3, Fig. 15). By using digital simulators and numerical method such as finite element analysis [37], an optimized structure can be iteratively developed in the future, fraction of structure weight and the gross take-off weight of the multicopter will be decreased significantly.

The methodology was also developed to practically predict the performance and support the conceptual sizing of an agriculture multicopter. In this specific mission profile, the mass of payload is decreased over time due to the amount of pesticide which is sprayed. Therefore, batteries requirement will be changed to adapt to and become suitable for the mission; moreover, the gross take-off weight of the vehicle will be decreased significantly. As a result, the cost of the product will decrease; transportation and operation costs will also be smaller. Battery capacity and its configuration have shown that they have a profound impact on electric UAV in general and multicopter in particular, as shown in Fig. 14.

Evaluation of the effect of input parameters, flight conditions on the GTOW from the algorithm is shown in Fig. 16. As seen in Fig. 16, for 10 kg of payload mass and the number of motors is eight, the flight time and load factor increased almost linearly with the GTOW. The propellers diameter, the number of motors and the number of cells for battery have great influence on the estimated multicopter's weight. Thus, designers can perform sizing

**Fig. 15.** Hover flight test for the 10 kg payload designed Agriculture Quadcopter. With the constraint of only used 80% of battery capacity to ensure its life, the hover time from the test is approximately 13 min.**Table 4**

Results for pesticide spraying multicopter with 15 minutes flight time: 10 minutes of spraying and 5 minutes of hovering.

		GTOW estimated (g)	Capacity required estimated (mAh)
Flow rate	1 L/min	19956	9215
Spraying speed	8 m/s		
Flow rate	1 L/min	20900	11827
Spraying speed	0 m/s (Hover)		
Flow rate	0 L/min (No spraying)	22994	16558
Spraying speed	0 m/s (Hover)		

and find the optimal multicopter configurations to meet mission requirements.

7. The influence of operating parameters over the quadcopter's weight

The algorithm then implemented with the calculation of payload mass changes over time and forward flight phase. The parasitic drag force applied on the multicopter during operation and free fall are handled by separating the vehicle to multiple element components such as arms, legs, body, and bottles; it is calculated analytically by the drag coefficient of components shapes as shown in Fig. 17.

As seen on the results shown in Table 4 and Fig. 18, the battery is the component whose mass influences the Agriculture Quadcopter's gross take-off weight the most by the mission requirement. Hovering and keeping the full load all time keeps the quadcopter's energy required at the largest rate comparing to the cases of payload weight reduced through time in hover or forward flight. By that, the battery's capacity could be adjusted to assure those mission requirements are met, but the total mass of the Quadcopters could still be kept in minimum.

In Fig. 18, the most significant fraction of a heavy payload Quadcopter is its payload for all of three cases. The second major portion of a multicopter is its structure, which is kept at constant at 20% total mass in the calculation; this fraction could be decreased by using structure optimization methods and lighter materials. The mass of propellers and ESCs have the smallest and second least percentage of mass distribution, respectively. With regards to the increase of the power requirement for three different kinds of missions, the ratio of battery mass is escalated greatly while the rate of motor weight is unnoticeable.

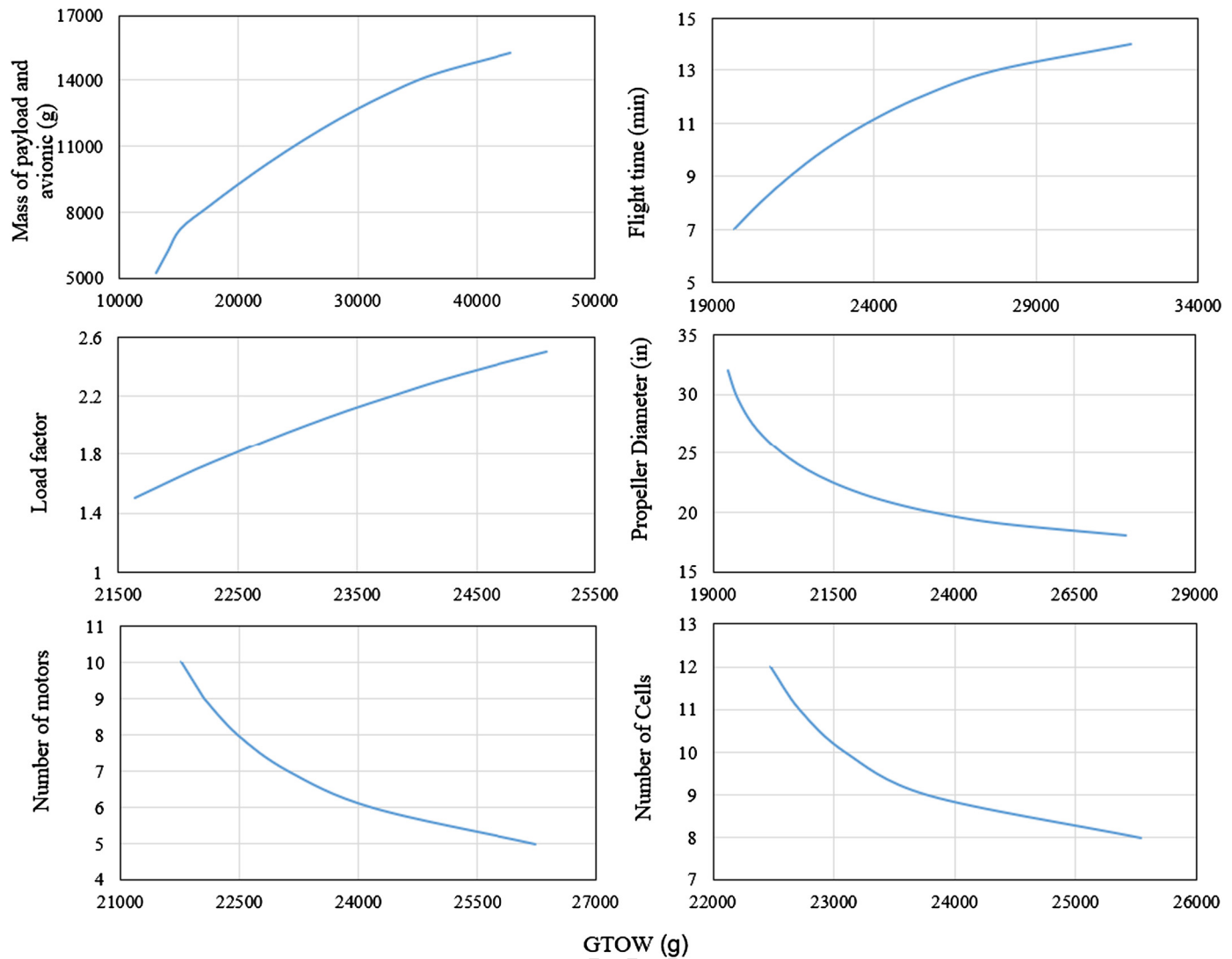


Fig. 16. The effect of input parameters to multicopters GTOW from the algorithm.

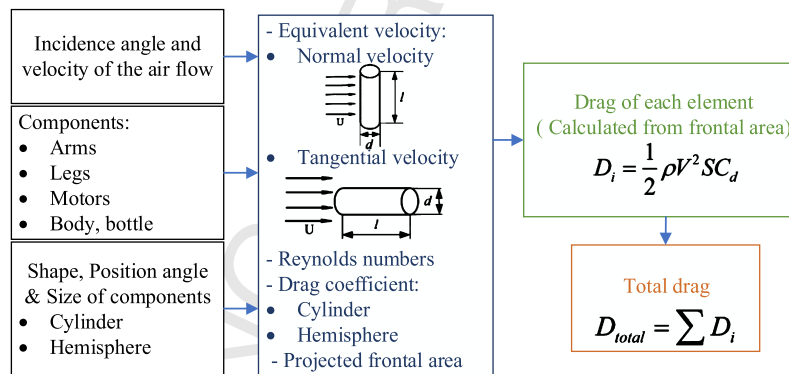


Fig. 17. Simple parasite drag calculation.

8. Conclusion

An algorithm for multicopters propulsion system sizing is introduced with the predicted gross take of weight within $\pm 4\%$ difference when comparing with an existing one. For the performance parameters, the differences between the calculated results and from manufacturers are minor (within 13%) for both cases (the Agras MG-1 and the designed Agriculture Quadcopter). As

we studied specifications brought by manufacturers and experimental data, the weight of motors, ESCs, batteries, propellers are expressed as functions of their characteristic parameters. By simplifying the electric propulsion system, parameterizing components and applying BEMT, this methodology is able to predict required RPM, power, motor current, component weights, battery capacity and also provides the sensitivity of design parameters to GTOW. Other factors such as max thrust over weight ratio, payload mass

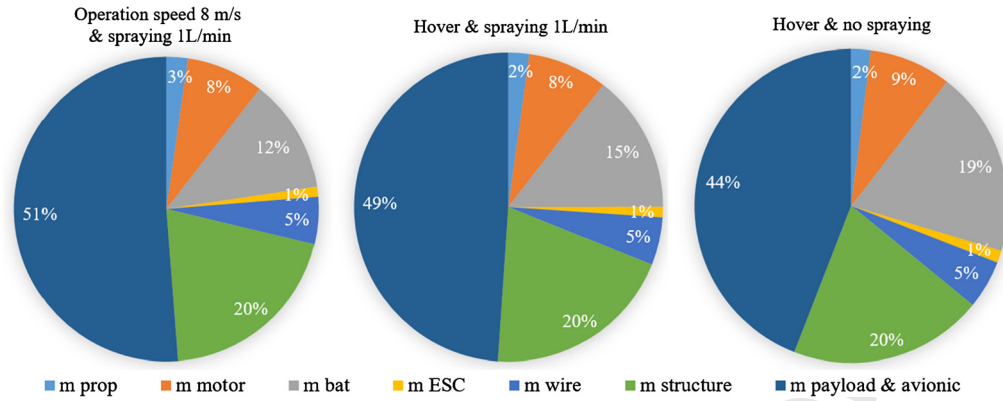


Fig. 18. Mass distribution for 10 kg payload pesticide spraying drone depends on mission profiles.

changes through time are accounted and it shows that they have a great impact on performance and total take-off weight of a multicopter. For specific mission profiles, a multicopter could be designed to obtain minimum take-off weight based on performance requirement such as the drop of payload with or without forward flight condition. When considering realistic mission profiles with forward flight stage and the decreasing of payload through time accounted, it shows that the multicopter's GTOW is significantly smaller than just hovering. An Agriculture Quadcopter is designed and constructed in reality by using the developed algorithm. The 10 kg payload pesticide sprayer is built from the methodology's results and does not show much difference. The outcome of hover time depends on batteries configuration which is given, the algorithm estimates the hover duration of the multicopter quite accurately when compared with field test results. In conclusion, this methodology has been proven that it is a good process to support the designing of multicopters.

Declaration of Competing Interest

None declared.

Acknowledgement

This work was supported by National Foundation for Science and Technology Development (NAFOSTED) of Vietnam (Project No. 107.01-2017.15).

Appendix

Momentum theory (Fig. 19)

By the conservation of mass, the mass flow rate is:

$$\dot{m} = \iint_{\infty} \rho \vec{V} d\vec{S} = \iint_2 \rho \vec{V} d\vec{S}$$

where $d\vec{S}$ is the outward pointing normal from the control volume. Therefore:

$$\dot{m} = \rho A_{\infty}(V_c + w) = \rho A(V_c + v_i)$$

Applying the principle of conservation of momentum:

$$-\vec{F} = \vec{T} = \iint_{\infty} \rho(\vec{V} d\vec{S}) \vec{V} - \iint_0 \rho(\vec{V} d\vec{S}) \vec{V}$$

In a steady climb, the velocity far upstream of the rotor is finite so that both terms on the right-hand side of the above equation are nonzero. Therefore, in this case:

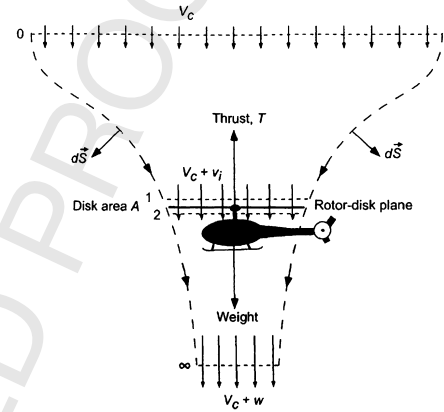


Fig. 19. Flow model for momentum theory analysis of a rotor in axial climbing flight [36].

$$T = \dot{m}(V_c + w) - \dot{m}V_c = \dot{m}w$$

The work done by the climbing rotor is now $T(V_c + v_i)$, then:

$$\begin{aligned} T(V_c + v_i) &= \iint_{\infty} \frac{1}{2} \rho(\vec{V} \cdot d\vec{S}) \vec{V}^2 - \iint_0 \frac{1}{2} \rho(\vec{V} \cdot d\vec{S}) \vec{V}^2 \\ &= \frac{1}{2} \dot{m}(V_c + w)^2 - \frac{1}{2} \dot{m}V_c^2 = \frac{1}{2} \dot{m}w(2V_c + w) \end{aligned}$$

For the climbing rotor, assuming that $w = 2v_i$:

$$T = \dot{m}w = \rho A(V_c + v_i)w_i = 2\rho A(V_c + v_i)v_i$$

So that:

$$\frac{T}{2A\rho} = (V_c + v_i)v_i$$

Computing the incremental thrust on the rotor annulus as the product of the mass flow rate through the annulus and twice the induced velocity at that section. In this case, the mass flow rate over the annulus of the disk is

$$d\dot{m} = \rho \cdot dA(V_c + v_i) = 2\pi\rho(V_c + v_i)y \cdot dy$$

The incremental thrust on the annulus because of the change in flow momentum along a stream-tube:

$$dT = 2\rho(V_c + v_i)v_i dA = 4\pi\rho(V_c + v_i)v_i y dy$$

The thrust coefficient is:

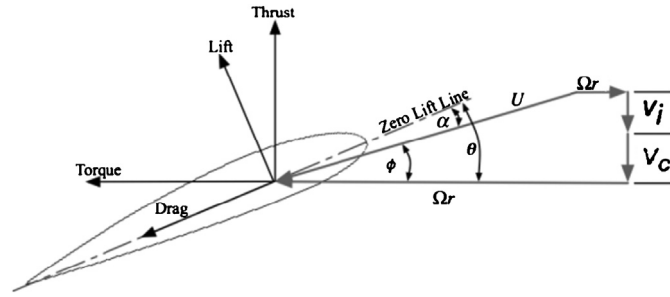


Fig. 20. Blade element's vector diagram.

$$dC_T = \frac{dT}{\rho(\pi R^2)(\Omega R)^2} = \frac{2\rho(V_c + v_i)v_i dA}{\rho\pi R^2(\Omega R)^2}$$

$$= \frac{2\rho(V_c + v_i)v_i(2\pi y \cdot dy)}{\rho\pi R^2(\Omega R)^2} = 4 \frac{(V_c + v_i)}{\Omega R} \left(\frac{v_i}{\Omega R} \right) \left(\frac{y}{R} \right) d \left(\frac{y}{R} \right)$$

$$= 4\lambda \lambda_i r dr = 4\lambda(\lambda - \lambda_c) r dr$$

The tip-hub loss effect can be incorporated into the BEMT (blade element momentum theory). Following, the modified equation using F of Prandtl tip-hub loss factor now is:

$$dC_T = 4\lambda F(\lambda - \lambda_c) r dr$$

Blade element theory (Fig. 20)

By using the fundamental assumption which determines the force, torque and power consuming on differential element throughout the blade length because of lift, drag characteristics of section profile of airfoil. At each element, the airfoil and its angle of attack involving to the geometry of blade, a differential propeller thrust, torque and power can be determined by:

$$dT = N_b(dL \cdot \cos \phi - dD \cdot \sin \phi)$$

$$\text{with } dL = \frac{1}{2}\rho U^2 c_L dy \text{ and } dD = \frac{1}{2}\rho U^2 c_D dy$$

$$U_P = v_c + v_i; \quad U_T = \Omega y; \quad U = \sqrt{U_T^2 + U_P^2}$$

and

$$\tan \phi = \frac{U_P}{U_T}$$

The inflow ratio λ is defined as:

$$\lambda = \frac{V_c + v_i}{\Omega R} = \frac{V_c + v_i}{\Omega y} \left(\frac{\Omega y}{\Omega R} \right) = \frac{U_P}{U_T} \left(\frac{y}{R} \right) = \phi \cdot r$$

The thrust coefficient is:

$$dC_T = \frac{dT}{\rho A(\Omega R)^2} = \frac{N_b(dL \cdot \cos \phi - dD \cdot \sin \phi)}{\rho(\pi R^2)(\Omega R)^2}$$

$$= \frac{N_b \frac{1}{2} \rho U^2 c (C_L \cdot \cos \phi - C_D \cdot \sin \phi) dy}{\rho(\pi R^2)(\Omega R)^2}$$

$$= \frac{1}{2} \sigma \frac{(C_L \cdot \cos \phi - C_D \cdot \sin \phi) (U_T^2 + U_P^2) dr}{(\Omega R)^2}$$

$$= \frac{1}{2} \sigma (C_L \cdot \cos \phi - C_D \cdot \sin \phi) (\lambda^2 + r^2) dr$$

Combining both momentum theory and BET, a list of relationship can be developed to determine the performance of propeller. By equating the thrust calculated by blade element theory and momentum theory for an annular element at radial position.

$$dT_{\text{momentum}} = dT_{\text{blade-element}}$$

$$\Leftrightarrow 4\lambda F(\lambda - \lambda_c) r dr = \frac{1}{2} \sigma (C_L \cdot \cos \phi - C_D \cdot \sin \phi) (\lambda^2 + r^2) dr$$

$$\Leftrightarrow \lambda^2 \left(4F - \frac{\sigma}{2r} (C_L(r_n) \cos \phi - C_D(r_n) \sin \phi) \right) - 4\lambda \lambda_c F r_n - \frac{1}{2} \sigma (C_L(r_n) \cos \phi - C_D(r_n) \sin \phi) r = 0$$

The inflow ratio λ from the quadratic equation:

$$\lambda(r_n) = \frac{\left(r_n \left(4F \lambda_c + \left(\frac{-(C_L(r_n) \cos \phi(r_n) - C_D(r_n) \sin \phi(r_n))^3 \sigma^2}{+8r_n(C_L(r_n) \cos \phi(r_n) - C_D(r_n) \sin \phi(r_n))^2 F \sigma + 16F^2 \lambda_c^2} \right)^{1/2} \right) \right)}{(8F r_n - (C_L(r_n) \cos \phi(r_n) - C_D(r_n) \sin \phi(r_n)) \sigma)}$$

References

- [1] M. Abarca, C. Saito, A. Angulo, J.A. Paredes, F. Cuellar, Design and development of an hexacopter for air quality monitoring at high altitudes, in: 13th IEEE Conf. Autom. Sci. Eng., 2017, pp. 1457–1462.
- [2] S. Anweiler, D. Piwowarski, Multicopter platform prototype for environmental monitoring, J. Clean. Prod. 155 (2017) 204–211, <https://doi.org/10.1016/j.jclepro.2016.10.132>.
- [3] C.C. Chang, J.L. Wang, C.Y. Chang, M.C. Liang, M.R. Lin, Development of a multicopter-carried whole air sampling apparatus and its applications in environmental studies, Chemosphere 144 (2016) 484–492, <https://doi.org/10.1016/j.chemosphere.2015.08.028>.
- [4] S. D'Oleire-Oltmanns, I. Marzolf, K.D. Peter, J.B. Ries, Unmanned aerial vehicle (UAV) for monitoring soil erosion in Morocco, Remote Sens. 4 (2012) 3390–3416, <https://doi.org/10.3390/rs4113390>.
- [5] A. Vale, R. Ventura, P. Carvalho, Application of unmanned aerial vehicles for radiological inspection, Fusion Eng. Des. 124 (2017) 492–495, <https://doi.org/10.1016/j.fusengdes.2017.06.002>.
- [6] J. Verbeke, D. Hulens, H. Ramon, T. Goedeme, J. De Schutter, The design and construction of a high endurance hexacopter suited for narrow corridors, in: 2014 Int. Conf. Unmanned Aircr. Syst. ICUAS 2014 – Conf. Proc., 2014, pp. 543–551.
- [7] A. Mahmood, Y. Kim, Leader-following formation control of quadcopters with heading, Aerosp. Sci. Technol. 47 (2015) 68–74, <https://doi.org/10.1016/j.ast.2015.09.009>.
- [8] Y. Yang, Y. Yan, Attitude regulation for unmanned quadrotors using adaptive fuzzy gain-scheduling sliding mode control, Aerosp. Sci. Technol. 1 (2016) 1–10, <https://doi.org/10.1016/j.ast.2016.04.005>.
- [9] S. Kiribayashi, J. Ashizawa, K. Nagatani, Modeling and design of tether powered multicopter, in: SSR 2015 - 2015 IEEE Int. Symp. Safety, Secur. Rescue Robot., 2016.
- [10] A. Bin Junaid, Y. Lee, Y. Kim, Design and implementation of autonomous wireless charging station for rotary-wing UAVs, Aerosp. Sci. Technol. 54 (2016) 253–266, <https://doi.org/10.1016/j.ast.2016.04.023>.
- [11] M.C. Achtelik, J. Stumpf, D. Gurdan, K.M. Doth, Design of a flexible high performance quadcopter platform breaking the MAV endurance record with laser power beaming, in: IEEE Int. Conf. Intell. Robot. Syst., 2011, pp. 5166–5172.
- [12] M. Achtelik, K.-M. Doth, D. Gurdan, J. Stumpf, Design of a multi rotor MAV with regard to efficiency, dynamics and redundancy, in: AIAA Guid. Navig. Control Conf., 2012, pp. 1–17.
- [13] B.Y. Suprpto, U. Dian, U. Dian, J. Muliadi, Design and development of heavy-lift hexacopter for heavy payload, in: Isemantic 2017, 2017, pp. 242–247.
- [14] xcopterCalc - Multicopter Calculator, www.ecalc.ch/xcopterCalc.php.
- [15] D. Shi, X. Dai, X. Zhang, Q. Quan, A practical performance evaluation method for electric multicopters, in: IEEE/ASME Trans. Mechatronics, 2017, pp. 1337–1348.
- [16] M. Kim, H. Joo, B. Jang, Conceptual multicopter sizing and performance analysis via component database, in: Int. Conf. Ubiquitous Futur. Networks, ICUFN, 2017, pp. 105–109.
- [17] D. Bershadsky, S. Haviland, E.N. Johnson, Electric multirotor UAV propulsion system sizing for performance prediction and design optimization, in: 57th AIAA/ASCE/AHS/ASC Struct. Struct. Dyn. Mater. Conf., 2016, pp. 1–22.
- [18] J.M. Winslow, V. Hrishikeshavan, I. Chopra, Design methodology for small scale unmanned quadrotors, in: 55th AIAA Aerosp. Sci. Meet., 2017.
- [19] S. Haviland, D. Bershadsky, D. Magree, E. Johnson, Development of a 500 gram vision-based autonomous quadrotor vehicle capable of indoor navigation Georgia Institute of Technology, in: Am. Helicopter Soc. 71st Annu. Forum, 2015.
- [20] C. Ampatis, E. Papadopoulos, Parametric design and optimization of multi-rotor aerial vehicles, Appl. Math. Inform. Sci. Eng. 30 (2014) 6266–6271, https://doi.org/10.1007/978-3-319-04720-1_1.

- [21] V.M. Arellano-Quintana, E.A. Portilla-Flores, E.A. Merchan-Cruz, P.A. Niño-Suarez, Multicopter design optimization using a genetic algorithm, in: Int. Conf. Unmanned Aircr. Syst., 2016, pp. 1313–1318.
- [22] M. Gatti, F. Giulietti, Preliminary design analysis methodology for electric multicopter, in: IFAC Proc. Vol., IFAC, 2013, pp. 58–63.
- [23] O. Gur, A. Rosen, Optimizing electric propulsion systems for unmanned aerial vehicles, *J. Aircr.* 46 (2009) 1340–1353, <https://doi.org/10.2514/1.41027>.
- [24] Ø. Magnussen, G. Hovland, M. Ottestad, Multicopter UAV design optimization, in: MESA 2014 - 10th IEEE/ASME Int. Conf. Mechatron. Embed. Syst. Appl. Conf. Proc., 2014.
- [25] M. Tyan, N. Van Nguyen, S. Kim, J.W. Lee, Comprehensive preliminary sizing/resizing method for a fixed wing – VTOL electric UAV, *Aerosp. Sci. Technol.* 71 (2017) 30–41, <https://doi.org/10.1016/j.ast.2017.09.008>.
- [26] Drive Calculator, www.drivecalc.de/.
- [27] A.M. Harrington, C. Kroninger, Characterization of Small DC Brushed and Brushless Motors, Aberdeen Proving Ground, USA, 2013.
- [28] M. Hassanalian, A. Abdelkefi, Classifications, applications, and design challenges of drones: a review, *Prog. Aerosp. Sci.* 91 (2017) 99–131, <https://doi.org/10.1016/j.paerosci.2017.04.003>.
- [29] Drone Batteries, <https://www.genstattu.com/tattu-uav-battery.html>.
- [30] I. Buchmann, BU-302: Series and Parallel Battery Configurations.
- [31] UIUC propeller data site, Univ. Illinois Urbana-Champaign.
- [32] Carbon fiber propellers, https://hobbyking.com/en_us/workbench/hardware-accessories/propellers.html?global_material=5056%0D.
- [33] N.A. Vu, J.W. Lee, Aerodynamic design optimization of helicopter rotor blades including airfoil shape for hover performance, *Chin. J. Aeronaut.* 26 (2013) 1–8, <https://doi.org/10.1016/j.cja.2012.12.008>.
- [34] N.A. Vu, J.W. Lee, Aerodynamic design optimization of helicopter rotor blades including airfoil shape for forward flight, *Aerosp. Sci. Technol.* 42 (2015) 106–117, <https://doi.org/10.1016/j.ast.2014.10.020>.
- [35] M. Imiela, High-fidelity optimization framework for helicopter rotors, *Aerosp. Sci. Technol.* 23 (2012) 2–16, <https://doi.org/10.1016/j.ast.2011.12.011>.
- [36] D. Leishman, Principles of Helicopter Aerodynamics, Cambridge University Press, New York, USA, 2006.
- [37] DJI, MG-1 SPECS, <https://www.dji.com/mg-1/info>. (Accessed 20 July 2007).
- [38] S. Kumar, P.C. Mishra, Finite element modeling for structural strength of quadcopter type multi mode vehicle, *Aerosp. Sci. Technol.* 53 (2016) 252–266, <https://doi.org/10.1016/j.ast.2016.03.020>.